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Code No: 123BP

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B.Tech II Year I Semester Examinations, November/December - 2017
DATA STRUCTURES
(Common to CSE, IT)

Time: 3 Hours **Max. Marks: 75**

Note: This question paper contains two parts A and B.
Part A is compulsory which carries 25 marks. Answer all questions in Part A.
Part B consists of 5 Units. Answer any one full question from each unit. Each question carries 10 marks and may have a, b, c as sub questions.

PART- A

(25 Marks)

1.a)	Define and List types of Arrays.	[2]
b)	Explain types of Linked Lists.	[3]
c)	Discuss about recursion and its implementation.	[2]
d)	Explain the ADT for Deque.	[3]
e)	Write about Properties of Binary Trees.	[2]
f)	Write about Directed & Undirected graphs.	[3]
g)	Write the C logic for Binary Search.	[2]
h)	Explain in brief about Insertion Sort.	[3]
i)	Write short note on Red Black Tree.	[2]
j)	Write the C Logic for Knuth-Morris-Pratt algorithm.	[3]

PART-B

(50 Marks)

2.a)	Write about Asymptotic Notations.	
b)	Discuss about linked representation of Sparse Matrix.	[5+5]
	OR	
3.a)	What is Performance Analysis? Explain the Techniques.	
b)	Explain about Singly Linked Lists Operations.	[5+5]
4.a)	Explain Evaluation of Postfix expression using Stack.	
b)	Write a program to implement Queue Operations.	[5+5]
	OR	
5.a)	Discuss about Circular queues operations.	
b)	Write a C program to implement stack using singly linked list.	[5+5]
6.a)	Describe Binary Tree traversals with examples.	
b)	Write a C Program to implement DFS.	[5+5]
	OR	
7.a)	Explain about Graph Representation methods.	
b)	Explain the Binary Tree operations with an example.	[5+5]

8.a) Write a C Program for Linear Search.
b) Explain Collision Resolution Methods. [5+5]

OR

9.a) Differentiate between Linear and Binary Search Methods.
b) Write a program for Quick Sort. [5+5]

10.a) Explain about AVL Tree with an example.
b) Give the examples for types of Tries. [5+5]

OR

11.a) Write about Splay Tree with an example.
b) Discuss about Binary Search Tree operations. [5+5]

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